# Co-Extrusion Blown Film Line
# Technical Modelling Report

*Spreadsheet Analysis · PDE/ODE System · State Space Model · Model Order Reduction · MPC Design*

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| **Document Type** | Technical Report |
| **Subject** | Process Modelling & Control |
| **System** | Co-Extrusion Blown Film Line |
| **Date** | July 28, 2026 |

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# Chapter 1

# Spreadsheet Analysis & Data Dictionary

## 1.1 Overview

This spreadsheet constitutes a comprehensive **data dictionary and tag mapping document** for an industrial co-extrusion blown film manufacturing line. It is used for SCADA/PLC data integration, mapping machine data points to a dataset for process monitoring, control, and data export. The data tags are primarily in German, with English translations provided in supplementary sheets.

**Document Purpose**

The spreadsheet serves as the **master data dictionary** for a co-extrusion blown film production line's data acquisition system. It maps every PLC/SCADA tag to a human-readable description, confirms all tags are included in the data export pipeline, and provides unit metadata — making it a critical reference for process engineers, data scientists, and system integrators.

## 1.2 Spreadsheet Structure

The spreadsheet contains four distinct sections:

1. **Station–Subsystem–DataPoint Mapping**
2. **Dataset Column Name → Export Flag**
3. **Clean Description ↔ Original Description**
4. **Category – Unit – Device Group**

### 1.2.1 Sheet 1: Station–Subsystem–DataPoint Mapping

Table 1.1: Column Descriptions — Sheet 1

| Column | Type | Description |
|---|---|---|
| Station | Integer | Numeric ID for the machine station (e.g., 0, 110, 112, 113, 114) |
| Subsystem | String | Functional area of the machine |
| DataPoint | String | Specific variable/tag being monitored |

Table 1.2: Stations and Subsystems Identified

| Station | Subsystem | Purpose |
|---|---|---|
| 0 | Production_Management | Production order tracking (`ActAuftrag`) |
| 110 | Extruders | Core extrusion process data (`Ex[0]`–`Ex[3]`) |
| 110 | Die_Head | Die/head heating zone control (`Extr[0]`) |
| 110 | Extrusion_General | Blower/fan load data (`Geblaese`) |
| 110 | Cooling_IBC | Internal Bubble Cooling system data |
| 110 | Dosing | Hopper/feeder pre-selection (`Trichter`) |
| 112 | Haul_Off_Tower | Film haul-off speed and position |
| 113 | Winder_1 | First winding station parameters |
| 114 | Winder_2 | Second winding station parameters |

### 1.2.2 Key DataPoint Categories

Table 1.3: Key DataPoint Categories for Extruders

| Tag Prefix | Description |
|---|---|
| `Dos[n]` | Dosing/feeder component data: actual proportion, rotation speed, throughput, material, bulk density, setpoints, valve status |
| `HeizungZone[n]` | Heating zone data: actual effect power, configuration, controller X/Y, setpoint temperature |
| `Foerderrate` | Feed/conveying rate |
| `MischDicht` | Mix density |
| `SDickeIst/Soll` | Layer thickness (actual, setpoint, percentage) |
| `SollLM / SollDS` | Setpoint values |
| `DissipationPwr` | Specific energy dissipation |
| `Massetemperatur` | Melt temperature |
| `druck[n].IstP` | Melt pressure actual |

### 1.2.3 Sheet 2: Dataset Column Name → Export Flag

Table 1.4: Export Flag Sheet Structure

| Column | Description |
|---|---|
| Dataset_Column_Name | Flattened/encoded variable name, e.g. `ST110_VAREx_1_Dos_2_IstAnteil` |
| In_Current_Data_Export | Whether the tag is included in the active data export (all marked "Yes") |

**Key Insight**

Every data point from Stations 110, 112, 113, and 114 is flagged as **"Yes"** — meaning all variables are actively exported to the dataset. The naming convention follows: `ST[Station]_VAR[Subsystem]_[Index]_[Parameter]`.

### 1.2.4 Sheet 3: Clean Description ↔ Original Description

Table 1.5: Example Description Mappings

| Clean Description | Meaning |
|---|---|
| Extruder running Job length setpoint (m) weight actual | Running metre weight output |
| Extruder A components 1 actual proportion | % share of material component 1 in extruder A |
| Die heating zone 3 set | Temperature setpoint for die zone 3 |
| Winder 1 remaining length current roll | Film remaining on current roll at winder 1 |
| Pellet blower 1 load | Load on pellet conveying blower 1 |

### 1.2.5 Sheet 4: Category – Unit – Device Group

Table 1.6: Unit Sheet Column Descriptions

| Column | Description |
|---|---|
| Category | Type of data point: Actual, Setpoint, Measurement, Command_Status, Drive_Status, Counter, Temperature |
| Unit_Metric | Metric unit (e.g., °C, kg/h, rpm, bar, m, %, kW, g/m, kWh/kg) |
| Unit_Imperial | Imperial equivalent (e.g., °F, lb/hr, psi, ft, lb/rpm) |
| Device_Group | Equipment group: extruder A/B/C, die zones 3–9, fan, blower, hopper, winder 1/2, air cooler, etc. |

## 1.3 System Description

This spreadsheet monitors a **multi-layer co-extrusion blown film line** comprising the following subsystems:

### 1.3.1 Extruders (A, B, C)

Up to four extruders, each with up to six dosing components. Parameters monitored include feed rates, rotation speeds, bulk density, material selection, layer thickness, melt temperature, melt pressure, energy dissipation, and eight heating barrel zones per extruder.

### 1.3.2 Die Head

Heating zones 3–9 on the central multi-layer die head, with actual temperature, setpoint, regulation ratio, and configuration monitored.

### 1.3.3 Cooling – IBC (Internal Bubble Cooling)

Three IBC units with calculated and visual speed setpoints, and three cooling devices with actual and setpoint temperatures.

### 1.3.4 Extrusion General / Blowers

Three pellet blowers with load monitoring, supply/exhaust air blowers, air ring blowers, and chill units.

### 1.3.5 Dosing / Hoppers

Twelve hoppers (`Trichter` 1–18, non-contiguous) with pre-selection settings.

### 1.3.6 Haul-Off Tower (Station 112)

Haul-off speed (setpoint & actual), input/output tracking.

### 1.3.7 Winders 1 & 2 (Stations 113 & 114)

Roll diameter, length (actual, previous, remaining, target), tension control, speed, winder drum parameters, maintenance counters, and roll change management.

## 1.4 Summary Statistics

Table 1.7: Spreadsheet Summary Statistics

| **Attribute** | **Detail** |
|---|---|
| Total Stations | 5 (0, 110, 112, 113, 114) |
| Subsystems | ≈10 |
| Extruders Monitored | 4 (Ex[0]–Ex[3] / A, B, C) |
| Dosing Components | Up to 6 per extruder |
| Heating Zones | Up to 8 barrel zones + die zones 3–9 |
| All Tags Exported | Yes (100%) |
| Language | German tags, English descriptions provided |
| Unit Systems | Both Metric & Imperial |

# Chapter 2

# System of PDEs/ODEs

## 2.1 Overview

A comprehensive mathematical framework is derived covering all major physical domains: extrusion, die head, cooling, haul-off, and winding.

## 2.2 Extruder Dynamics

### 2.2.1 Screw & Melt Flow (Mass Balance per Extruder $k$)

The volumetric flow rate through extruder $k$ is governed by the screw geometry and rotational speed:

**Extruder Flow Rate**

$$Q_k = \alpha_k N_k - \beta_k \frac{\Delta P_k}{\mu_k(T_k, \dot{\gamma}_k)} \tag{2.1}$$

where:

- $Q_k$ — volumetric flow rate of extruder $k$ ($m^3/s$)
- $\alpha_k$ — drag flow coefficient (screw geometry)
- $N_k$ — screw rotational speed (rpm)
- $\beta_k$ — pressure flow coefficient
- $\Delta P_k$ — pressure drop across the screw
- $\mu_k$ — non-Newtonian melt viscosity

### 2.2.2 Non-Newtonian Viscosity (Power-Law Model)

$$\mu_k(T_k, \dot{\gamma}_k) = m_k(T_k) \cdot \dot{\gamma}_k^{n_k - 1} \tag{2.2}$$

with temperature dependence via the Arrhenius relation:

$$m_k(T_k) = m_{0,k} \exp\left(\frac{E_{a,k}}{RT_k}\right) \tag{2.3}$$

where $\dot{\gamma}_k$ is the shear rate ($s^{-1}$), $n_k$ is the power-law index, $E_{a,k}$ is the activation energy (J/mol), and $R$ is the universal gas constant.

### 2.2.3 Melt Pressure ODE

$$\frac{dP_k}{dt} = \frac{K_k}{\rho_k} \left(\dot{m}_{in,k} - \dot{m}_{out,k}\right) \tag{2.4}$$

where $K_k$ is the bulk modulus of the melt.

### 2.2.4 Energy Balance PDE (along screw axial direction $z$)

**Extruder Energy Balance PDE**

$$\rho_k c_{p,k} \frac{\partial T_k}{\partial t} + \rho_k c_{p,k} v_{z,k} \frac{\partial T_k}{\partial z} = \lambda_k \frac{\partial^2 T_k}{\partial z^2} + \eta_k \dot{\gamma}_k^2 + \dot{q}_{wall,k}(z,t) \tag{2.5}$$

where:

- $v_{z,k}$ — axial melt velocity
- $\lambda_k$ — thermal conductivity of melt
- $\eta_k \dot{\gamma}_k^2$ — viscous dissipation (linked to `DissipationPwr`)
- $\dot{q}_{wall,k}$ — heat flux from barrel heating zones

### 2.2.5 Barrel Zone Heat Flux (per zone $j$, extruder $k$)

$$\dot{q}_{wall,k,j} = h_{k,j} \left(T_{set,k,j} - T_k(z_j,t)\right) \cdot P_{eff,k,j}(t) \tag{2.6}$$

$$\frac{dT_{set,k,j}}{dt} = \frac{1}{\tau_{k,j}} \left(T_{sp,k,j} - T_{set,k,j}\right) - K_{P,k,j}\, e_{k,j}(t) \tag{2.7}$$

where $e_{k,j} = T_{set,k,j} - T_{act,k,j}$ is the PID error signal (linked to `Regler_X`, `Regler_Y`).

## 2.3 Dosing / Feeder Dynamics (Component $i$, Extruder $k$)

### 2.3.1 Component Mass Balance ODE

$$\frac{dm_{i,k}}{dt} = \dot{m}_{feed,i,k} - \dot{m}_{out,i,k} \tag{2.8}$$

$$\dot{m}_{out,i,k} = \rho_{bulk,i,k} \cdot Q_{dos,i,k}(N_{dos,i,k}) \tag{2.9}$$

### 2.3.2 Actual Proportion

$$\phi_{i,k}(t) = \frac{\dot{m}_{out,i,k}}{\sum_{i=1}^{n} \dot{m}_{out,i,k}} \tag{2.10}$$

### 2.3.3 Dosing Control Law (PI)

$$\frac{dN_{dos,i,k}}{dt} = K_{c,i,k}\left(\phi_{i,k}^{set} - \phi_{i,k}(t)\right) + \frac{K_{c,i,k}}{\tau_{I,i,k}} \int_0^t \left(\phi_{i,k}^{set} - \phi_{i,k}\right) d\tau \tag{2.11}$$

### 2.3.4 Mix Density

$$\rho_{mix,k}(t) = \sum_{i=1}^{n} \phi_{i,k}(t) \cdot \rho_{bulk,i,k} \tag{2.12}$$

## 2.4 Die Head Thermal Dynamics

### 2.4.1 Die Zone Temperature PDE (radial $r$, axial $z$)

**Die Head PDE**

$$\rho_{die} c_{p,die} \frac{\partial T_{die}}{\partial t} = \lambda_{die}\left(\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T_{die}}{\partial r}\right) + \frac{\partial^2 T_{die}}{\partial z^2}\right) + \dot{q}_{heater}(r, z, t) \tag{2.13}$$

### 2.4.2 Heater Power Control per Zone $j$

$$P_{eff,j}(t) = \frac{P_{eff,j}^{max}}{100} \cdot u_j(t), \quad u_j \in [0,\, 100]\,\% \tag{2.14}$$

$$\frac{du_j}{dt} = K_{P,j}\frac{de_j}{dt} + \frac{K_{P,j}}{\tau_{I,j}} e_j(t) \tag{2.15}$$

where $e_j = T_{sp,j} - T_{act,j}$ (linked to `SollTemp`, `Regler_Y`, `Regler_X`).

### 2.4.3 Melt Flow in Die (Annular Channel)

$$\frac{\partial v_z}{\partial t} = -\frac{1}{\rho}\frac{\partial P}{\partial z} + \frac{1}{r}\frac{\partial}{\partial r}\left(r\mu\frac{\partial v_z}{\partial r}\right) \tag{2.16}$$

with boundary conditions: $v_z(R_{inner}, t) = 0, \quad v_z(R_{outer}, t) = 0.$

## 2.5 Blown Film Bubble Dynamics

### 2.5.1 Bubble Radius PDE

$$\frac{\partial R}{\partial t} + v_z\frac{\partial R}{\partial z} = \frac{R}{2}\left(\dot{\varepsilon}_\theta - \dot{\varepsilon}_z\right) \tag{2.17}$$

### 2.5.2 Film Thickness Evolution

**Film Thickness PDE**

$$\frac{\partial h}{\partial t} + v_z \frac{\partial h}{\partial z} = -h\left(\dot{\varepsilon}_z + \dot{\varepsilon}_\theta\right) \tag{2.18}$$

where $h(z,t)$ is the local film thickness (linked to `SDickeIst`), and $\dot{\varepsilon}_z$, $\dot{\varepsilon}_\theta$ are the axial and hoop strain rates.

### 2.5.3 Axial Force Balance

$$F_z = 2\pi R h \sigma_{zz} + \pi R^2 \Delta P_{bubble} \tag{2.19}$$

$$\frac{dF_z}{dz} = 0 \quad \text{(quasi-steady axial tension)} \tag{2.20}$$

### 2.5.4 Bubble Internal Pressure (IBC Control)

$$\frac{dP_{bubble}}{dt} = \frac{\gamma_{air}}{V_{bubble}}\left(\dot{m}_{in,IBC} - \dot{m}_{out,IBC}\right) \cdot \frac{R_{gas} T_{air}}{M_{air}} \tag{2.21}$$

## 2.6 Cooling Dynamics (IBC + Air Ring)

### 2.6.1 Bubble Film Temperature PDE

$$\rho_f c_{p,f} h \frac{\partial T_f}{\partial t} = -h_{conv,ext}\left(T_f - T_{air,ext}\right) - h_{conv,int}\left(T_f - T_{IBC}\right) + \dot{q}_{rad} \tag{2.22}$$

### 2.6.2 IBC Air Temperature ODE (unit $l$)

$$\rho_{air} V_{IBC,l} c_{p,air} \frac{dT_{IBC,l}}{dt} = \dot{m}_{in,l} c_{p,air} T_{in,l} - \dot{m}_{out,l} c_{p,air} T_{IBC,l} - UA_l\left(T_{IBC,l} - T_{film}\right) \tag{2.23}$$

### 2.6.3 IBC Speed Control ODE

$$\frac{dN_{IBC,l}}{dt} = K_{IBC}\left(N^{set}_{IBC,l} - N_{IBC,l}\right) \tag{2.24}$$

### 2.6.4 Cooling Device Temperature ODE

$$M_{cool,l} c_{p,cool} \frac{dT_{cool,l}}{dt} = \dot{Q}_{in,l} - UA_{cool,l}\left(T_{cool,l} - T_{amb}\right) \tag{2.25}$$

## 2.7 Haul-Off Tower Dynamics

### 2.7.1 Film Velocity and Tension

$$\frac{dv_{haul}}{dt} = \frac{1}{J_{haul}}\left(F_{drive} - F_{tension} - F_{friction}\right) \tag{2.26}$$

### 2.7.2 Film Tension ODE

$$\frac{d\sigma_{haul}}{dt} = \frac{E_f \cdot h \cdot W}{L_{haul}} (v_{haul} - v_{bubble,FLH}) \tag{2.27}$$

### 2.7.3 Speed Control

$$\frac{dv_{haul}}{dt} = K_v \left(v_{haul}^{set} - v_{haul}\right) \tag{2.28}$$

## 2.8 Winder Dynamics (Winder $w = 1, 2$)

### 2.8.1 Roll Build-Up ODE

$$\frac{dR_{roll,w}}{dt} = \frac{h_{film} \cdot v_{haul}}{2\pi R_{roll,w} \cdot \eta_{pack}} \tag{2.29}$$

### 2.8.2 Accumulated Length

$$\frac{dL_w}{dt} = v_{haul}(t) \qquad L_{remaining,w}(t) = L_{target,w} - L_w(t) \tag{2.30}$$

### 2.8.3 Winder Drum Speed

$$\omega_{drum,w}(t) = \frac{v_{haul}(t)}{R_{roll,w}(t)} \tag{2.31}$$

### 2.8.4 Web Tension Control ODE

$$J_w \frac{d\omega_{drum,w}}{dt} = T_{drive,w} - T_{tension,w} - T_{friction,w} \tag{2.32}$$

$$T_{tension,w} = \sigma_{web,w} \cdot h_{film} \cdot W_{film} \tag{2.33}$$

### 2.8.5 Tension Taper (Clp Curve)

$$\sigma_{web,w}^{set}(R) = \sigma_{0,w} \left(1 - \kappa_w \cdot \frac{R_{roll,w} - R_{core,w}}{R_{max,w} - R_{core,w}}\right) \tag{2.34}$$

where $\kappa_w$ is the taper factor (linked to `ClpTens`, `WdSpTens`).

## 2.9 Layer Thickness & Output Coupling

### 2.9.1 Total Output

$$\dot{m}_{total}(t) = \sum_{k=0}^{K-1} \rho_{mix,k}(t) \cdot Q_k(t) \tag{2.35}$$

### 2.9.2 Layer Thickness Fraction

$$\delta_k(t) = \frac{Q_k(t)}{\sum_j Q_j(t)} \cdot h_{total}(t) \tag{2.36}$$

### 2.9.3 Global Film Thickness

**Global Film Thickness**

$$h_{total}(z,t) = \frac{\dot{m}_{total}}{2\pi R(z,t) \cdot \rho_{mix} \cdot v_z(z,t)} \tag{2.37}$$

## 2.10 Production Management

$$\frac{dL_{job}}{dt} = v_{haul}(t) \qquad \frac{dm_{job}}{dt} = \rho_{film} \cdot h_{total} \cdot W_{film} \cdot v_{haul}(t) \tag{2.38}$$

# Chapter 3

# State Space Model

## 3.1 General Form

The nonlinear state space representation is:

**Nonlinear State Space**

$$\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t),\, \mathbf{u}(t),\, t) \tag{3.1}$$
$$\mathbf{y}(t) = \mathbf{g}(\mathbf{x}(t),\, \mathbf{u}(t),\, t) \tag{3.2}$$

Linearised around an operating point $(\mathbf{x}_0, \mathbf{u}_0)$:

**Linearised State Space**

$$\dot{\tilde{\mathbf{x}}} = \mathbf{A}\tilde{\mathbf{x}} + \mathbf{B}\tilde{\mathbf{u}} \tag{3.3}$$
$$\tilde{\mathbf{y}} = \mathbf{C}\tilde{\mathbf{x}} + \mathbf{D}\tilde{\mathbf{u}} \tag{3.4}$$

where $\tilde{\mathbf{x}} = \mathbf{x} - \mathbf{x}_0$, $\tilde{\mathbf{u}} = \mathbf{u} - \mathbf{u}_0$.

## 3.2 State Vector Definition

The full state vector is partitioned by subsystem:

$$\mathbf{x}(t) = \begin{bmatrix} \mathbf{x}_{ext} \\ \mathbf{x}_{dos} \\ \mathbf{x}_{die} \\ \mathbf{x}_{bub} \\ \mathbf{x}_{cool} \\ \mathbf{x}_{haul} \\ \mathbf{x}_{wind} \end{bmatrix} \in \mathbb{R}^{n_x} \tag{3.5}$$

### 3.2.1 Extruder States (per extruder $k$, zone $j$)

$$\mathbf{x}_{ext,k} = \begin{bmatrix} T_{k,1} \\ T_{k,2} \\ \vdots \\ T_{k,8} \\ P_k \\ \dot{m}_{out,k} \\ T_{melt,k} \end{bmatrix} \in \mathbb{R}^{11}, \qquad \mathbf{x}_{ext} = \begin{bmatrix} \mathbf{x}_{ext,0}^\top & \mathbf{x}_{ext,1}^\top & \mathbf{x}_{ext,2}^\top & \mathbf{x}_{ext,3}^\top \end{bmatrix}^\top \in \mathbb{R}^{44} \tag{3.6}$$

### 3.2.2 Dosing States

$$\mathbf{x}_{dos,i,k} = \begin{bmatrix} \phi_{i,k} \\ N_{dos,i,k} \\ m_{i,k} \end{bmatrix} \in \mathbb{R}^{3}, \qquad \mathbf{x}_{dos} \in \mathbb{R}^{60} \quad \text{(4 extruders} \times \text{5 components)} \tag{3.7}$$

### 3.2.3 Die Head States

$$\mathbf{x}_{die} = \begin{bmatrix} T_{die,3} \\ \vdots \\ T_{die,9} \\ u_{die,3} \\ \vdots \\ u_{die,9} \end{bmatrix} \in \mathbb{R}^{14} \tag{3.8}$$

### 3.2.4 Bubble States

$$\mathbf{x}_{bub} = \begin{bmatrix} R_{bub} \\ h_{film} \\ P_{bub} \\ v_{z,bub} \\ T_{film} \\ \sigma_{zz} \end{bmatrix} \in \mathbb{R}^{6} \tag{3.9}$$

### 3.2.5 Cooling States

$$\mathbf{x}_{cool} = \begin{bmatrix} T_{IBC,1} \\ T_{IBC,2} \\ T_{IBC,3} \\ N_{IBC,1} \\ N_{IBC,2} \\ N_{IBC,3} \\ T_{cool,1} \\ T_{cool,2} \\ T_{cool,3} \end{bmatrix} \in \mathbb{R}^{9} \tag{3.10}$$

### 3.2.6 Haul-Off States

$$\mathbf{x}_{haul} = \begin{bmatrix} v_{haul} \\ \sigma_{haul} \\ L_{job} \end{bmatrix} \in \mathbb{R}^3 \tag{3.11}$$

### 3.2.7 Winder States

$$\mathbf{x}_{wind,w} = \begin{bmatrix} \omega_{drum,w} \\ R_{roll,w} \\ L_w \\ \sigma_{web,w} \\ T_{drive,w} \end{bmatrix} \in \mathbb{R}^5, \qquad \mathbf{x}_{wind} \in \mathbb{R}^{10} \tag{3.12}$$

### 3.2.8 Total State Dimension

**Total State Dimension**

$$n_x = 44 + 60 + 14 + 6 + 9 + 3 + 10 = \mathbf{146} \tag{3.13}$$

## 3.3 Input Vector

Table 3.1: Input Vector Summary

| Group | Symbol | Variables | Dim |
|---|---|---|---|
| Extruder | $\mathbf{u}_{ext}$ | Screw speeds $N_k$, zone temp setpoints $T_{sp,k,j}$ | $4 \times 9 = 36$ |
| Dosing | $\mathbf{u}_{dos}$ | Dosing speed setpoints, valve states | $4 \times 5 \times 2 = 40$ |
| Die Head | $\mathbf{u}_{die}$ | Die zone temp setpoints $T_{sp,die,j}$ | 7 |
| Cooling | $\mathbf{u}_{cool}$ | IBC speed setpoints, cooling device setpoints | 6 |
| Haul-Off | $\mathbf{u}_{haul}$ | Haul-off speed setpoint $v_{haul}^{set}$ | 1 |
| Winder | $\mathbf{u}_{wind}$ | Winder speed, tension setpoint, taper | 6 |
| | | **Total** | **96** |

## 3.4 Output Vector

Table 3.2: Output Vector Summary ($n_y = 58$)

| Output | Description | Source Tag |
|---|---|---|
| $T_{melt,k}$ | Melt temperature per extruder | `Massetemperatur` |
| $P_k$ | Melt pressure per extruder | `druck[n].IstP` |
| $\dot{m}_{out,k}$ | Mass flow rate per extruder | `Foerderrate` |
| $h_{film}$ | Layer thickness actual | `SDickeIst` |
| $\delta_k/h$ | Layer thickness % | `SDickeProz` |
| $R_{bub}$ | Bubble radius | (derived) |
| $N_{IBC,l}$ | IBC speeds actual | `IBC[l].Ist_n_Calc` |
| $T_{cool,l}$ | Cooling device temperatures | `KuehlGeraet[l].IstTemp` |
| $v_{haul}$ | Haul-off speed actual | `Abzug[1].IstZu` |
| $R_{roll,w}$ | Roll diameter | `DiaRollRc` |
| $\sigma_{web,w}$ | Web tension | `WdSpTens` |
| $L_{rem,w}$ | Remaining length | `RemainingLen` |
| $T_{die,j}$ | Die zone temperatures actual | `HeizungZone[j].SollTemp` |
| $\phi_{i,k}$ | Component proportions | `Dos[i].IstAnteil` |

## 3.5 Nonlinear State Equations

### 3.5.1 Extruder Thermal States

$$\dot{T}_{k,j} = \frac{1}{\rho_k c_{p,k} V_{k,j}} \Big[ h_{k,j} A_{k,j} (T_{sp,k,j} - T_{k,j}) P_{eff,k,j} - \dot{m}_{out,k} c_{p,k} (T_{k,j} - T_{k,j-1}) + \eta_k \dot{\gamma}_k^2 V_{k,j} \Big] \tag{3.14}$$

### 3.5.2 Melt Pressure State

$$\dot{P}_k = \frac{K_k}{\rho_k} \left( \rho_k \alpha_k N_k - \rho_k \beta_k \frac{P_k - P_{die}}{\mu_k} - \dot{m}_{out,k} \right) \tag{3.15}$$

### 3.5.3 Melt Temperature at Exit

$$\dot{T}_{melt,k} = \frac{1}{\tau_{melt,k}} \left( \bar{T}_k - T_{melt,k} \right) + \frac{\eta_k \dot{\gamma}_k^2}{\rho_k c_{p,k}}, \qquad \bar{T}_k = \frac{1}{8} \sum_{j=1}^{8} T_{k,j} \tag{3.16}$$

### 3.5.4 Dosing States

$$\dot{\phi}_{i,k} = \frac{1}{\tau_{dos}} \left( \frac{N_{dos,i,k} \rho_{bulk,i,k}}{\sum_i N_{dos,i,k} \rho_{bulk,i,k}} - \phi_{i,k} \right) \tag{3.17}$$

$$\dot{N}_{dos,i,k} = K_c \left( \phi_{i,k}^{set} - \phi_{i,k} \right) + \frac{K_c}{\tau_I} \int_0^t e_{dos} \, d\tau \tag{3.18}$$

$$\dot{m}_{i,k} = \dot{m}_{feed,i,k} - N_{dos,i,k} \rho_{bulk,i,k} \alpha_{dos,i} \tag{3.19}$$

### 3.5.5 Die Head States

$$\dot{T}_{die,j} = \frac{1}{\rho_{die} c_{p,die} V_{die,j}} \Big[ P_{eff,die,j} - h_{die,j} A_{die,j} (T_{die,j} - T_{melt,avg}) \Big] \tag{3.20}$$

$$\dot{u}_{die,j} = \frac{1}{\tau_{die,j}} (T_{sp,die,j} - T_{die,j}) \tag{3.21}$$

### 3.5.6 Bubble States

$$\dot{R}_{bub} = \tfrac{R_{bub}}{2} (\dot{\varepsilon}_\theta - \dot{\varepsilon}_z) \tag{3.22}$$

$$\dot{h}_{film} = -h_{film} (\dot{\varepsilon}_z + \dot{\varepsilon}_\theta) \tag{3.23}$$

$$\dot{P}_{bub} = \frac{\gamma_{air} R_{gas} T_{air}}{M_{air} V_{bub}} (\dot{m}_{in,IBC} - \dot{m}_{out,IBC}) \tag{3.24}$$

$$\dot{v}_{z,bub} = \frac{1}{\rho_f h_{film}} \left( \frac{\partial \sigma_{zz}}{\partial z} + \frac{P_{bub} - P_{atm}}{R_{bub}} \right) \tag{3.25}$$

$$\dot{T}_{film} = \frac{1}{\rho_f c_{p,f} h_{film}} \Big[ -h_{ext}(T_{film} - T_{air}) - h_{int}(T_{film} - T_{IBC,avg}) \Big] \tag{3.26}$$

$$\dot{\sigma}_{zz} = E_f \left( \dot{\varepsilon}_z - \frac{\sigma_{zz}}{E_f \tau_{relax}} \right) \tag{3.27}$$

### 3.5.7 Cooling States

$$\dot{T}_{IBC,l} = \frac{1}{\rho_{air} V_{IBC,l} c_{p,air}} \Big[ \dot{m}_{in,l} c_{p,air} (T_{in,l} - T_{IBC,l}) - UA_l (T_{IBC,l} - T_{film}) \Big] \tag{3.28}$$

$$\dot{N}_{IBC,l} = \frac{1}{\tau_{IBC}} \left( N_{IBC,l}^{set} - N_{IBC,l} \right) \tag{3.29}$$

$$\dot{T}_{cool,l} = \frac{1}{M_{cool,l} c_{p,cool}} \Big[ \dot{Q}_{in,l} - UA_{cool,l} (T_{cool,l} - T_{amb}) \Big] \tag{3.30}$$

### 3.5.8 Haul-Off States

$$\dot{v}_{haul} = \frac{1}{J_{haul}} \Big[ K_v (v_{haul}^{set} - v_{haul}) - \sigma_{haul} h_{film} W_{film} - F_{fric} \Big] \tag{3.31}$$

$$\dot{\sigma}_{haul} = \frac{E_f h_{film} W_{film}}{L_{haul}} (v_{haul} - v_{bub,exit}) \tag{3.32}$$

$$\dot{L}_{job} = v_{haul} \tag{3.33}$$

### 3.5.9 Winder States

$$\dot{\omega}_{drum,w} = \frac{1}{J_w}\Big[T_{drive,w} - \sigma_{web,w} h_{film} W_{film} R_{roll,w} - B_w \omega_{drum,w}\Big] \tag{3.34}$$

$$\dot{R}_{roll,w} = \frac{h_{film} v_{haul}}{2\pi R_{roll,w}} \tag{3.35}$$

$$\dot{L}_w = v_{haul} \tag{3.36}$$

$$\dot{\sigma}_{web,w} = \frac{E_f h_{film} W_{film}}{L_{span,w}} \left(\omega_{drum,w} R_{roll,w} - v_{haul}\right) \tag{3.37}$$

$$\dot{T}_{drive,w} = \frac{1}{\tau_{drive,w}} \left(T^{set}_{drive,w} - T_{drive,w}\right) \tag{3.38}$$

## 3.6 Linearised Matrices

$$\mathbf{A} = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{\mathbf{x}_0,\mathbf{u}_0} \in \mathbb{R}^{146\times 146}, \quad \mathbf{B} = \left.\frac{\partial \mathbf{f}}{\partial \mathbf{u}}\right|_{\mathbf{x}_0,\mathbf{u}_0} \in \mathbb{R}^{146\times 96} \tag{3.39}$$

$$\mathbf{C} = \left.\frac{\partial \mathbf{g}}{\partial \mathbf{x}}\right|_{\mathbf{x}_0,\mathbf{u}_0} \in \mathbb{R}^{58\times 146}, \quad \mathbf{D} = \left.\frac{\partial \mathbf{g}}{\partial \mathbf{u}}\right|_{\mathbf{x}_0,\mathbf{u}_0} \in \mathbb{R}^{58\times 96} \tag{3.40}$$

### 3.6.1 Block Structure of A

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_{ee} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{A}_{de} & \mathbf{A}_{dd} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{A}_{die,e} & \mathbf{0} & \mathbf{A}_{die} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{A}_{be} & \mathbf{0} & \mathbf{A}_{bd} & \mathbf{A}_{bb} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{cb} & \mathbf{A}_{cc} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{hb} & \mathbf{A}_{hc} & \mathbf{A}_{hh} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{A}_{wh} & \mathbf{A}_{ww} \end{bmatrix} \tag{3.41}$$

### 3.6.2 Controllability & Observability

$$\operatorname{rank}\Big[\mathbf{B} \quad \mathbf{AB} \quad \cdots \quad \mathbf{A}^{145}\mathbf{B}\Big] = 146 \qquad \text{(Controllability)} \tag{3.42}$$

$$\operatorname{rank}\begin{bmatrix} \mathbf{C} \\ \mathbf{CA} \\ \vdots \\ \mathbf{CA}^{145} \end{bmatrix} = 146 \qquad \text{(Observability)} \tag{3.43}$$

$$\operatorname{Re}(\lambda_i(\mathbf{A})) < 0, \quad \forall\, i = 1,\ldots,146 \qquad \text{(Stability)} \tag{3.44}$$

# Chapter 4

# Model Order Reduction & MPC Design

## 4.1 Reduction Pipeline Overview

**Reduction Pipeline**

$$\underbrace{\mathbf{x} \in \mathbb{R}^{146}}_{\text{Full Order}} \xrightarrow{\text{Singular Perturbation}} \underbrace{\mathbf{x}_s \in \mathbb{R}^{90}}_{\text{Slow States}} \xrightarrow{\text{Balanced Truncation}} \underbrace{\hat{\mathbf{x}} \in \mathbb{R}^{40}}_{\text{Intermediate}} \xrightarrow{\text{POD/Galerkin}} \underbrace{\tilde{\mathbf{x}} \in \mathbb{R}^{22}}_{\text{MPC-Ready}} \tag{4.1}$$

## 4.2 Stage 1: Singular Perturbation

### 4.2.1 Time-Scale Separation

Partition states into slow ($\tau > \tau_{MPC}$) and fast ($\tau \ll \tau_{MPC}$):

$$\begin{bmatrix} \dot{\mathbf{x}}_s \\ \varepsilon \dot{\mathbf{x}}_f \end{bmatrix} = \begin{bmatrix} \mathbf{A}_{ss} & \mathbf{A}_{sf} \\ \mathbf{A}_{fs} & \mathbf{A}_{ff} \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_f \end{bmatrix} + \begin{bmatrix} \mathbf{B}_s \\ \mathbf{B}_f \end{bmatrix} \mathbf{u} \tag{4.2}$$

Setting $\varepsilon \to 0$ (quasi-steady-state):

$$\mathbf{x}_f^* = -\mathbf{A}_{ff}^{-1} \left( \mathbf{A}_{fs} \mathbf{x}_s + \mathbf{B}_f \mathbf{u} \right) \tag{4.3}$$

Reduced slow subsystem:

$$\dot{\mathbf{x}}_s = \underbrace{\left( \mathbf{A}_{ss} - \mathbf{A}_{sf} \mathbf{A}_{ff}^{-1} \mathbf{A}_{fs} \right)}_{\mathbf{A}_s} \mathbf{x}_s + \underbrace{\left( \mathbf{B}_s - \mathbf{A}_{sf} \mathbf{A}_{ff}^{-1} \mathbf{B}_f \right)}_{\mathbf{B}_s} \mathbf{u} \tag{4.4}$$

### 4.2.2 Time Constant Classification

Table 4.1: State Time Constant Classification

| Subsystem | States | Typical $\tau$ (s) | Class |
|---|---|---|---|
| Barrel zone temps | $T_{k,j}$ | 200–600 | Slow |
| Melt temperature | $T_{melt,k}$ | 60–180 | Slow |
| Die zone temps | $T_{die,j}$ | 100–400 | Slow |
| Film thickness | $h_{film}$ | 30–90 | Slow |
| Bubble radius | $R_{bub}$ | 20–60 | Slow |
| Bubble pressure | $P_{bub}$ | 10–30 | Slow |
| Roll build-up | $R_{roll,w}$ | 300–1800 | Slow |
| Web tension | $\sigma_{web}$ | 5–20 | Slow |
| Melt pressure | $P_k$ | 1–5 | Fast |
| Dosing speed | $N_{dos}$ | 0.5–2 | Fast |
| IBC speed | $N_{IBC}$ | 1–3 | Fast |
| Drive torque | $T_{drive}$ | 0.1–1 | Fast |

## 4.3 Stage 2: Balanced Truncation

### 4.3.1 Gramian Computation

Solve the Lyapunov equations:

$$\mathbf{A}_s\mathbf{W}_c + \mathbf{W}_c\mathbf{A}_s^\top + \mathbf{B}_s\mathbf{B}_s^\top = \mathbf{0} \tag{4.5}$$
$$\mathbf{A}_s^\top\mathbf{W}_o + \mathbf{W}_o\mathbf{A}_s + \mathbf{C}_s^\top\mathbf{C}_s = \mathbf{0} \tag{4.6}$$

### 4.3.2 Balancing Transformation

Cholesky factorisations:

$$\mathbf{W}_c = \mathbf{L}_c\mathbf{L}_c^\top, \qquad \mathbf{W}_o = \mathbf{L}_o\mathbf{L}_o^\top \tag{4.7}$$

SVD of the cross-Gramian product:

$$\mathbf{L}_o^\top\mathbf{L}_c = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top, \qquad \mathbf{\Sigma} = \mathrm{diag}(\sigma_1, \sigma_2, \ldots, \sigma_{n_s}), \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{n_s} \geq 0 \tag{4.8}$$

Balancing transformation matrices:

$$\mathbf{T} = \mathbf{L}_c\mathbf{V}\mathbf{\Sigma}^{-1/2}, \qquad \mathbf{T}^{-1} = \mathbf{\Sigma}^{-1/2}\mathbf{U}^\top\mathbf{L}_o^\top \tag{4.9}$$

### 4.3.3 Truncation Order Selection

Retain $r_1$ states satisfying:

$$\frac{\sum_{i=1}^{r_1}\sigma_i}{\sum_{i=1}^{n_s}\sigma_i} \geq 1 - \epsilon_{BT}, \qquad \epsilon_{BT} = 0.01 \tag{4.10}$$

Guaranteed $\mathcal{H}_\infty$ error bound:

$$\|\mathbf{G}(s) - \mathbf{G}_{r_1}(s)\|_{\mathcal{H}_\infty} \leq 2 \sum_{i=r_1+1}^{n_s} \sigma_i \tag{4.11}$$

## 4.4 Stage 3: POD / Galerkin Projection

### 4.4.1 Snapshot Matrix

$$\mathbf{S} = \begin{bmatrix} \mathbf{x}_{r_1}(t_1) & \mathbf{x}_{r_1}(t_2) & \cdots & \mathbf{x}_{r_1}(t_N) \end{bmatrix} \in \mathbb{R}^{r_1 \times N} \tag{4.12}$$

### 4.4.2 POD Basis via SVD

$$\mathbf{S} = \mathbf{\Phi}\mathbf{\Lambda}\mathbf{\Psi}^\top \tag{4.13}$$

Retain $r$ modes capturing 99% of energy:

$$\frac{\sum_{i=1}^{r} \lambda_i^2}{\sum_{i=1}^{r_1} \lambda_i^2} \geq 0.99 \tag{4.14}$$

### 4.4.3 Galerkin Projection

$$\tilde{\mathbf{A}} = \mathbf{\Phi}_r^\top \hat{\mathbf{A}}_{11} \mathbf{\Phi}_r \in \mathbb{R}^{r \times r}, \quad \tilde{\mathbf{B}} = \mathbf{\Phi}_r^\top \hat{\mathbf{B}}_1 \in \mathbb{R}^{r \times n_u}, \quad \tilde{\mathbf{C}} = \hat{\mathbf{C}}_1 \mathbf{\Phi}_r \in \mathbb{R}^{n_y \times r} \tag{4.15}$$

## 4.5 Stage 4: Discretisation (ZOH)

$$\mathbf{A}_d = e^{\tilde{\mathbf{A}}T_s}, \qquad \mathbf{B}_d = \tilde{\mathbf{A}}^{-1}\left(e^{\tilde{\mathbf{A}}T_s} - \mathbf{I}\right)\tilde{\mathbf{B}}, \qquad T_s = 3\,\mathrm{s} \tag{4.16}$$

## 4.6 Stage 5: Augmented Model for Offset-Free MPC

$$\begin{bmatrix} \tilde{\mathbf{x}}_{k+1} \\ \mathbf{d}_{k+1} \end{bmatrix} = \underbrace{\begin{bmatrix} \mathbf{A}_d & \mathbf{B}_d^d \\ \mathbf{0} & \mathbf{I} \end{bmatrix}}_{\mathbf{A}_{aug}} \begin{bmatrix} \tilde{\mathbf{x}}_k \\ \mathbf{d}_k \end{bmatrix} + \underbrace{\begin{bmatrix} \mathbf{B}_d \\ \mathbf{0} \end{bmatrix}}_{\mathbf{B}_{aug}} \mathbf{u}_k \tag{4.17}$$

$$\mathbf{y}_k = \underbrace{\begin{bmatrix} \mathbf{C}_d & \mathbf{I} \end{bmatrix}}_{\mathbf{C}_{aug}} \begin{bmatrix} \tilde{\mathbf{x}}_k \\ \mathbf{d}_k \end{bmatrix} \tag{4.18}$$

## 4.7 Kalman Filter for State Estimation

$$\hat{\mathbf{x}}_{aug,k|k} = \hat{\mathbf{x}}_{aug,k|k-1} + \mathbf{K}_f\left(\mathbf{y}_k - \mathbf{C}_{aug}\hat{\mathbf{x}}_{aug,k|k-1}\right) \tag{4.19}$$

$$\mathbf{K}_f = \mathbf{P}_k \mathbf{C}_{aug}^\top \left(\mathbf{C}_{aug}\mathbf{P}_k\mathbf{C}_{aug}^\top + \mathbf{R}_n\right)^{-1} \tag{4.20}$$

$$\mathbf{P}_{k+1} = \mathbf{A}_{aug}\left(\mathbf{P}_k - \mathbf{K}_f\mathbf{C}_{aug}\mathbf{P}_k\right)\mathbf{A}_{aug}^\top + \mathbf{Q}_n \tag{4.21}$$

## 4.8 MPC Formulation

### 4.8.1 Prediction Model

$$\mathbf{Y} = \mathbf{F}\hat{\mathbf{x}}_{aug,k} + \mathbf{G}\Delta\mathbf{U} \tag{4.22}$$

$$\mathbf{F} = \begin{bmatrix} \mathbf{C}_{aug}\mathbf{A}_{aug} \\ \mathbf{C}_{aug}\mathbf{A}_{aug}^2 \\ \vdots \\ \mathbf{C}_{aug}\mathbf{A}_{aug}^{N_p} \end{bmatrix} \in \mathbb{R}^{N_p n_y \times (r+n_y)} \tag{4.23}$$

$$\mathbf{G} = \begin{bmatrix} \mathbf{C}_{aug}\mathbf{B}_{aug} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{C}_{aug}\mathbf{A}_{aug}\mathbf{B}_{aug} & \mathbf{C}_{aug}\mathbf{B}_{aug} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{C}_{aug}\mathbf{A}_{aug}^{N_p-1}\mathbf{B}_{aug} & \cdots & \cdots & \mathbf{C}_{aug}\mathbf{B}_{aug} \end{bmatrix} \in \mathbb{R}^{N_p n_y \times N_c n_u} \tag{4.24}$$

### 4.8.2 Optimisation Problem

**MPC Optimisation Problem**

$$\min_{\Delta\mathbf{U}} \quad J = \underbrace{\|\mathbf{Y} - \mathbf{Y}_{ref}\|^2_{\mathbf{Q}_{MPC}}}_{\text{tracking}} + \underbrace{\|\Delta\mathbf{U}\|^2_{\mathbf{R}_{MPC}}}_{\text{effort}} + \underbrace{\|\mathbf{U}\|^2_{\mathbf{S}}}_{\text{regularisation}} \tag{4.25}$$

subject to:

$$\mathbf{u}_{min} \le \mathbf{u}_k \le \mathbf{u}_{max} \tag{4.26}$$

$$\Delta\mathbf{u}_{min} \le \Delta\mathbf{u}_k \le \Delta\mathbf{u}_{max} \tag{4.27}$$

$$\mathbf{y}_{min} \le \mathbf{y}_k \le \mathbf{y}_{max} \tag{4.28}$$

### 4.8.3 Compact QP Form

$$\min_{\Delta\mathbf{U}} \quad \frac{1}{2}\Delta\mathbf{U}^\top\mathbf{H}\Delta\mathbf{U} + \mathbf{f}^\top\Delta\mathbf{U} \quad \text{s.t.} \quad \mathbf{M}\Delta\mathbf{U} \le \mathbf{b} \tag{4.29}$$

$$\mathbf{H} = 2\left(\mathbf{G}^\top\mathbf{Q}_{MPC}\mathbf{G} + \mathbf{R}_{MPC}\right) \in \mathbb{R}^{N_c n_u \times N_c n_u} \tag{4.30}$$

$$\mathbf{f} = 2\mathbf{G}^\top\mathbf{Q}_{MPC}\left(\mathbf{F}\hat{\mathbf{x}}_{aug} - \mathbf{Y}_{ref}\right) \tag{4.31}$$

### 4.8.4 MPC Tuning Parameters

Table 4.2: MPC Tuning Parameters

| Parameter | Symbol | Recommended Value |
|---|---|---|
| Prediction horizon | $N_p$ | 20–40 steps |
| Control horizon | $N_c$ | 5–10 steps |
| Sampling time | $T_s$ | 3–5 s |
| Output weight | $\mathbf{Q}_{MPC}$ | Diagonal, priority-based |
| Input weight | $\mathbf{R}_{MPC}$ | $0.01–0.1 \cdot \mathbf{I}$ |

### 4.8.5 Output Priority Weighting

$$\mathbf{Q}_{MPC} = \operatorname{diag}\underbrace{(q_h, q_h, q_h, q_h)}_{\text{thickness}} \oplus \underbrace{(q_T, \ldots, q_T)}_{\text{temperatures}} \oplus \underbrace{(q_\sigma, q_\sigma)}_{\text{tensions}} \oplus \cdots, \qquad q_h \gg q_T \gg q_\sigma \tag{4.32}$$

## 4.9 Final Reduced Model

**Discrete Reduced MPC Model**

$$\tilde{\mathbf{x}}_{k+1} = \mathbf{A}_d\tilde{\mathbf{x}}_k + \mathbf{B}_d\mathbf{u}_k + \mathbf{w}_k \tag{4.33}$$

$$\mathbf{y}_k = \mathbf{C}_d\tilde{\mathbf{x}}_k + \mathbf{v}_k \tag{4.34}$$

$$\tilde{\mathbf{x}} \in \mathbb{R}^r, \quad \mathbf{u} \in \mathbb{R}^{n_u}, \quad \mathbf{y} \in \mathbb{R}^{n_y}, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_n), \quad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_n) \tag{4.35}$$

## 4.10 Complete Reduction Summary

Table 4.3: Model Order Reduction Pipeline Summary

| Stage | Method | States | Error Bound |
|---|---|---|---|
| Full Model | — | 146 | — |
| Singular Perturbation | Time-scale separation | 90 | $\mathcal{O}(\varepsilon)$ |
| Balanced Truncation | Gramian / HSV | 40 | $2\sum_{i>r_1}\sigma_i$ |
| POD / Galerkin | SVD snapshots | 22 | $1-\frac{\sum_r \lambda_i^2}{\sum \lambda_i^2}$ |
| Discretisation (ZOH) | $T_s = 3\,\text{s}$ | 22 | $\mathcal{O}(T_s^2)$ |
| **MPC Model** | **Augmented** | $\mathbf{22+58=80}$ | **Certified** |